

Solvark Design System & Component Library Specification

SUMMARY

This document defines the reusable UI components, design tokens, interaction patterns, animation rules, accessibility standards, and 3D component architecture for Solvark.

PROJECT

Solvark Premium Website & Admin CMS

VERSION

1.0

DOCUMENT TYPE

Design System Specification (DSS)

PURPOSE

Every screen—marketing website, admin panel, future client portal, and future mobile apps—must use this design system to ensure consistency and scalability.

DESIGN PHILOSOPHY

Premium; Minimal; Engineering-focused; Modern; Motion-driven; 3D-enabled; Accessible; Fast. The UI should communicate precision engineering, not just attractive visuals.

DESIGN TOKENS

Colors: Neutral (Black 950, Gray 950, Gray 900, Gray 800, Gray 700, White); Brand (Primary, Primary Hover, Secondary, Accent); Semantic (Success, Warning, Error, Info); Surface (Glass, Elevated, Card, Modal, Sidebar).

TYPOGRAPHY SCALE

Hero XL 96px; Hero 72px; Display 60px; H1 48px; H2 40px; H3 32px; H4 24px; H5 20px; Body Large 18px; Body 16px; Small 14px; Caption 12px.

GRID SYSTEM

Desktop: 12 Columns; Tablet: 8 Columns; Mobile: 4 Columns; Container: 1440px.

COMPONENT CATEGORIES

Layout (AppShell, Container, Section, Grid, Stack, Sidebar, Header, Footer); Navigation (Navbar, Mega Menu, Breadcrumb, Tabs, Sidebar Navigation, Mobile Drawer, Command Menu); Buttons variants (Primary, Secondary, Glass, Ghost, Outline, Icon, Gradient, Danger), states (Default, Hover, Active, Loading, Disabled), animations (Magnetic Cursor, Glow, Ripple, Scale, Border Animation).

FORMS

Components: Input, Textarea, Select, Multi Select, Checkbox, Radio, Toggle, OTP Input, Password Field, Date Picker, File Upload, Image Upload, Video Upload, GLB Upload. Validation: Client, Server. Error: Inline. Success: Animated.

CARDS

Types: Glass Card, Project Card, Service Card, Team Card, Blog Card, Pricing Card, Statistic Card, Review Card. Effects: Hover, Lift, Glow, Rotate, Depth Shadow.

TABLES

Admin Components: Data Table, Pagination, Search, Filters, Bulk Actions, Export, Column Visibility, Row Selection.

MODALS

Types: Confirmation, Image Viewer, Video Viewer, Project Preview, Gallery Preview, Offer Preview, Form Dialog. Animations: Scale, Blur, Fade, Spring Motion.

DRAWERS

Mobile Navigation, Filters, Notifications, Settings, Quick Actions.

NOTIFICATIONS

Components: Toast, Alert, Snackbar, Banner. Variants: Success, Error, Warning, Info.

DASHBOARD WIDGETS

Statistics, Revenue, Charts, Visitors, Leads, Projects, Calendar, Recent Activity.

CHARTS

Supported: Line, Area, Bar, Pie, Donut, Radar, Heatmap. Interactive: Hover, Zoom, Export.

3D COMPONENT LIBRARY

Hero Scene (Floating Logo, Glass Shapes, Dynamic Lights, Particle System, Camera Motion); 3D Service Cards (GLB Model, Reflection, Rotation, Cursor Interaction); Portfolio Viewer (Rotate, Zoom, Fullscreen, Device Switching, Auto Rotation); Product Carousel (Infinite Scroll, Keyboard, Touch, Mouse Wheel); Globe (Client Locations, Animated Routes, Pulsing Pins, Auto Rotation); Background Elements (Floating Cubes, Abstract Rings, Code Particles, Wireframe Shapes, Glass Orbs).

GALLERY COMPONENTS

Gallery Grid (Masonry, Equal Grid, Horizontal, Fullscreen); Gallery Lightbox (Zoom, Pan, Next, Previous, Keyboard Navigation, Fullscreen, Download Control, Share); Before / After Slider (website redesigns, branding transformations, UI improvements); Video Gallery (MP4, WebM, YouTube, Vimeo).

ADVERTISEMENT COMPONENTS

Offer Card fields: Badge, Background, Price, CTA, Timer. Banner: Image, Video, 3D Model. Popup: Offer Modal, Newsletter, Exit Intent. Floating CTA: WhatsApp, Book Call, Quote Request.

ANIMATION STANDARDS

Entry: Fade, Scale, Blur, Slide. Hover: Lift, Rotate, Glow, Border Animation. Scroll: Reveal, Pin, Horizontal Scroll, Counter Animation, Parallax. Cursor: Magnetic Buttons, Spotlight, Cursor Glow, Trail. Page: Smooth Transition, Shared Element Animation.

ICONOGRAPHY

Primary Library: Lucide. 3D Icons: Custom GLB. Animated SVG: Lottie.

ILLUSTRATION STYLE

Use: Wireframes, Abstract Shapes, Technology Diagrams, Minimal Icons. Avoid: Cartoon illustrations, Generic stock vectors.

IMAGE STANDARDS

Formats: AVIF, WebP. Fallback: JPG. Optimization: Automatic. Lazy Loading: Required.

VIDEO STANDARDS

Autoplay: Muted, Loop, Lazy Loaded. Poster Image: Required.

ACCESSIBILITY

Every component must support Keyboard, Screen Reader, Focus Ring, Reduced Motion, High Contrast, ARIA Labels.

PERFORMANCE RULES

Maximum: Initial JS Bundle <250 KB (excluding lazy-loaded assets), GLB Model <2 MB, Texture <1024px when practical. Lazy Load: Required. Dynamic Import: Required. Suspense: Required.

CMS COMPONENTS

Homepage Builder (Hero Editor, CTA Editor, Statistics, Services, Offers, Gallery, Testimonials, FAQ); Media Manager (Images, Videos, 3D Models, HDR Files); Offer Manager (Discounts, Campaigns, Timers, Display Rules).

ADMIN COMPONENTS

Dashboard, Projects, Clients, Reviews, Gallery, Offers, SEO, Analytics, Media, Settings, Roles, Permissions. Every module includes Search, Filters, Bulk Actions, Import/Export, Activity History.

COMPONENT STATES

Default, Hover, Focus, Active, Loading, Disabled, Empty, Error, Success, Skeleton.

MOTION PERFORMANCE

Target: 60 FPS on modern desktops; Graceful degradation on low-end devices; Respect prefers-reduced-motion; Pause heavy animations when off-screen; WebGL fallback for unsupported browsers.

FILE ORGANIZATION

/components /layout /navigation /buttons /cards /forms /tables /dialogs /charts /gallery /3d /animations /dashboard /cms /feedback

COMPONENT DOCUMENTATION REQUIREMENTS

Every component must include Purpose, Props, Variants, States, Accessibility notes, Usage examples, Performance considerations.

DESIGN QA CHECKLIST

Pixel-perfect spacing; Consistent typography; Responsive behavior; Animation timing consistency; Keyboard accessibility; Dark theme correctness; Image optimization; 3D performance; CMS editability; Lighthouse scores ≥ 95 .

FUTURE COMPONENT ROADMAP

AI Assistant Panel; Interactive ROI Calculator; Pricing Configurator; Client Project Timeline; Live Collaboration Widgets; 3D Product Configurator; WebXR/AR Showcase; Theme Customizer; Plugin Marketplace.

DESIGN SYSTEM COMPLETION CRITERIA

The design system is complete when every reusable interface element exists as a documented component; marketing site and admin panel share the same visual language; all 3D interactions follow consistent motion and lighting rules; components are reusable, accessible, performant, and fully documented; future modules can be built without introducing new visual patterns unless intentionally extending the design system.

RECOMMENDED NEXT DOCUMENT

Frontend Architecture & Implementation Guide, covering component hierarchy, routing, state management, animation architecture, React Three Fiber scene organization, CMS integration, and coding standards for developers.